SOUMENDRA GANGULY

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SKILLS

C, Python, Linux	14+ yrs
POSIX shell, Bash	12+ yrs
*NIX sys/net admin	12+ yrs
SageMath	10+ yrs
SymPy, NumPy	10+ yrs
LaTeX	14+ yrs
Lua	3+ yrs
GNU Emacs	11+ yrs
HTML, JavaScript	10+ yrs
CSS, Bootstrap	10+ yrs
Git, Docker	8+ yrs
CI/CD (GitHub Actions)	5+ yrs
Apache, NGINX	8+ yrs
Django, Flask	8+ yrs
Hugo, Jinja	3+ yrs
Lark parser	1.5 yrs
OpenAI API	2+ yrs
PyTorch	2+ mth
Stable Diffusion	2+ mth

PROJECTS

- github.com/8vasu/sg-diffusion
  PyTorch, Stable Diffusion, Docker, CUDA
- github.com/8vasu/2windows.el
  GNU Emacs, ace-window

EDUCATION

2019–2023, Texas A&M University
PhD, Mathematics

2017–2019, Clemson University
MS, Mathematical Sciences

2014–2016, Chennai Math. Inst.
MSc, Mathematics

2011–2014, Chennai Math. Inst.
BSc, Math and Computer Sci.

LANGUAGES

English, Hindi, Bangla ■■■■■
Danish (DU3, Module 3) ■■■

PROFILE

I am an experienced computer scientist and programmer with over fourteen years of expertise in software development, system administration, and web technologies. I am highly proficient in Linux and other *nix platforms. I have contributed significant features, implemented unit tests, and fixed bugs in several major open-source projects.

In addition to my technical background, I am a mathematician with a strong research focus on the analytic behavior of automorphic L-functions and number theory.

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher in Mathematics at Aarhus University — Sep 2023–NOW
Research

- **Problem solving, analytical reasoning, creativity, initiative:** Employed rigorous proof techniques to analyze and solve complex, abstract problems in the theory of automorphic L-functions, contributing to the understanding of their underlying structures.
- **Leadership, project management:** I am the principal organizer and driving force behind a 2-week international event: The Aarhus Automorphic Forms Conference and Summer School, August 4–15, Aarhus University. This involves managing a team of 9 organizers, coordinating 21 invited speakers, 12 contributed talk speakers, overseeing the logistics of the event including accommodation, transportation, dining, visa applications, and managing budget, time, and funding applications for 100+ participants from around the world.
- **Data analysis and communication:** Interpretation and presentation of data and research results to small and large groups. My academic CV is available on my website linked in the header.
- **Research:** Performed literature review and analysis of existing research papers. Wrote a convenient command-line arXiv preprint batch downloader: github.com/8vasu/paper.py.

Independent programmer — Dec 2023–NOW
Collaboration with SymPy

- **Feature development:** Added LaTeX matrix and complex number parsing capability to the official **SymPy** project in a significant pull request.
- **Package development:** Developed github.com/8vasu/compuTeX, a GNU Emacs package enabling users to input complex mathematical expressions in LaTeX format and output the evaluated expression back in LaTeX with a single keystroke, powered by SymPy.
- **Tools:** Python, SymPy, the Lark parser with EBNF grammar, regular expressions, LaTeX , GNU Emacs, Emacs Lisp, Git.

Independent programmer — Sep 2020–NOW
Collaboration with CPython, FreeBSD, NetBSD, and util-linux

- **Feature development, long-term sustained contribution:**
  - Added 7 new functions to the *os*, *termios*, *tty*, and *pty* modules of the official **CPython** project in an ongoing GitHub issue containing 20 pull requests over 5+ years, significantly enhancing the Python standard library's functionality.
  - Added 2 functions to the standard C library of the **FreeBSD** operating system and wrote the manual page for them.
  - Improved the standard *script(1)* and *scriptreplay(1)* programs on FreeBSD, NetBSD, and util-linux as demonstration of these functionalities.
- **Bug fixing:** Addressed and resolved pseudoterminal-related issues within the CPython project.
- **Software testing:** Created comprehensive **unit tests** to validate the reliability and correctness of newly introduced features.
- **Package development:** Developed github.com/8vasu/stty.py for clean, Pythonic POSIX *stty(1)*-style terminal manipulation. Published on PyPI via automated GitHub Actions workflows.
- **Tools:** C, Python, POSIX/BSD/GNU termios and pseudoterminal APIs, unit testing, Git, Docker, GNU Autotools, *nroff(1)*, *script(1)*, *scriptreplay(1)*, *stty(1)*, JSON, GitHub Actions.